

Ahmet Neçirvan Doğan

Istanbul, Türkiye | andogan22@ku.edu.tr | +90 531 508 66 18
linkedin.com/in/ahmet-necirvan-dogan | github.com/ahmetnecirvandogan

Education

Koç University, B.S. in Computer Engineering – Istanbul, Türkiye Oct 2022 – Jun 2027
Track: Artificial Intelligence & Data Analytics | Minor: Media and Visual Arts

Experience

- Undergraduate Researcher**, Intelligent User Interfaces Lab, Koç University Feb 2026 – Present
- Developing AI-driven, sketch-based interfaces for neural cloth reconstruction as part of a joint Computer Graphics and HCI research initiative between Koç University and the University of Manchester. Co-advised by Prof. Metin Sezgin and Dr. Zahra Montazeri; paper targeted for submission to Eurographics 2027.
- Software Engineer Intern**, Ministry of Energy and Natural Resources – Türkiye Aug 2026 – Present
- Developing a 3D digital twin and solar panel simulation in Unity (C#), covering scene construction, simulation logic, and real-time 3D visualization.
- Software Engineer Intern**, SimTek Simulation Technologies – Ankara, Türkiye Jul 2025 – Aug 2025
- Developed features for a graphics-intensive simulation product in C++ and OpenGL, working inside the core engineering team's codebase and code review process.
- Cast Member**, Walt Disney World (International College Program) – Florida, USA Jun 2024 – Sep 2024
- Selected for Disney's international program; operated a high-volume attraction within a multinational team, coordinating daily throughput and guest safety in a fully English-speaking environment.

Projects

- Sketch Generator** – Python, 3D Simulation & Rendering Mar 2026 – Present
- Designed a multi-stage pipeline that simulates 3D cloth draping and renders photorealistic samples with ground-truth depth and normal maps, producing training-ready datasets optimized for sketch-conditioned diffusion models.
- Clash Royale with JavaFX** – Java 11, JavaFX 21, Maven Oct 2025 – Jan 2026
- Built a real-time strategy card game with single-player AI, local and network multiplayer, custom deck building, and an arena editor; implemented A* pathfinding, state-machine unit behavior, and client-server state synchronization.
- Interactive World Building Simulation** – C++, OpenGL, GLSL Apr 2025 – Jun 2025
- Engineered a real-time 3D sandbox where users construct and manipulate a dynamic virtual world on a modular grid, featuring Phong shading with dynamic shadows, fluid simulation for rivers, and a day-night cycle.

Technical Skills

Languages: C++, Python, Java, C#, MATLAB, GLSL
Graphics & Simulation: OpenGL, Unity, Shader Programming, Blender, GLFW, GLEW
Machine Learning & Vision: PyTorch, NumPy, OpenCV
Tools: Git, GitHub, Linux, Maven, JavaFX, LaTeX, Claude Code

Volunteering

- Volunteer**, Make-A-Wish Türkiye Aug 2026 – Present
- Supporting wish experiences for children with critical illnesses.

Certifications

- AI Fluency: Framework and Foundations** – Anthropic, Jul 2026
- Software Engineering Specialization** – The Hong Kong University of Science and Technology, Jul 2025
- Graph Search, Shortest Paths, and Data Structures** – Stanford University, Jan 2025